

Key Concepts in Cloud Computing

1. Introduction to Cloud Computing

Cloud computing is the delivery of computing services — including servers, storage, databases, networking, software, and analytics — over the internet (the cloud). Instead of owning and maintaining physical hardware, users access resources on demand from cloud providers and pay only for what they use.

- **Key Idea:** Resources are shared, scalable, and accessible from anywhere via the internet.
- **Examples of Cloud Providers:** Amazon Web Services (AWS), Microsoft Azure, Google Cloud Platform (GCP).

2. Essential Characteristics of Cloud Computing

NIST defines five essential characteristics of cloud computing:

- **On-Demand Self-Service:** Users can provision resources automatically without human interaction from the provider.
- **Broad Network Access:** Services are available over the network and accessible via standard devices (laptops, phones, tablets).
- **Resource Pooling:** Provider's resources are shared among multiple customers using a multi-tenant model. Resources are dynamically assigned based on demand.
- **Rapid Elasticity:** Resources can be scaled up or down quickly, appearing unlimited to the user.
- **Measured Service:** Resource usage is monitored, controlled, and billed automatically (pay-as-you-go model).

3. Cloud Service Models

Cloud services are offered at three main levels, often visualized as a stack:

IaaS (Infrastructure as a Service)

- Provides virtualized computing infrastructure: virtual machines, storage, networking.
- User manages OS, middleware, and applications. Provider manages physical hardware.
- **Examples:** AWS EC2, Google Compute Engine, Microsoft Azure VMs.
- **Use case:** Hosting websites, running custom applications, backup and recovery.

PaaS (Platform as a Service)

- Provides a platform with tools and environment to develop, test, and deploy applications.
- User manages only the application and data. Provider manages OS, runtime, and infrastructure.
- **Examples:** Google App Engine, AWS Elastic Beanstalk, Microsoft Azure App Service.

- **Use case:** Application development without managing underlying infrastructure.

SaaS (Software as a Service)

- Delivers fully functional software applications over the internet.
- User only uses the application. Provider manages everything underneath.
- **Examples:** Gmail, Google Docs, Salesforce, Microsoft Office 365, Dropbox.
- **Use case:** Email, collaboration tools, CRM systems.

4. Cloud Deployment Models

- **Public Cloud:** Resources are owned and operated by a third-party provider and shared over the internet. Cost-effective and scalable. Example: AWS, Azure.
- **Private Cloud:** Cloud infrastructure is used exclusively by a single organization. Hosted on-premises or by a third party. More control and security.
- **Hybrid Cloud:** A combination of public and private clouds. Allows data and applications to move between them. Balances flexibility and security.
- **Community Cloud:** Shared infrastructure for a specific community with common concerns (e.g., government agencies, healthcare organizations).
- **Multi-Cloud:** Using services from more than one cloud provider to avoid vendor lock-in and improve resilience.

5. Virtualization

Virtualization is the foundation of cloud computing. It allows a single physical machine to run multiple virtual machines (VMs), each with its own OS and resources.

- **Hypervisor:** Software layer that creates and manages VMs. Two types: Type 1 (bare-metal, runs directly on hardware) and Type 2 (hosted, runs on top of an OS).
- **Virtual Machine (VM):** An isolated software simulation of a physical computer.
- **Containers:** Lightweight alternative to VMs. Share the host OS kernel. Faster and more resource-efficient. Example: Docker, Kubernetes.
- **Benefits:** Resource efficiency, isolation, easier management, fast provisioning.

6. Cloud Storage

Cloud storage allows data to be stored on remote servers accessed via the internet. Types include:

- **Block Storage:** Stores data in fixed-size blocks. Used for databases and VMs. Example: AWS EBS.
- **Object Storage:** Stores data as objects with metadata. Best for unstructured data like images and videos. Example: AWS S3, Google Cloud Storage.
- **File Storage:** Hierarchical file system storage accessible over a network. Example: AWS EFS.
- **Key Properties:** Durability, availability, scalability, and data redundancy through replication.

7. Cloud Security

Security is a critical concern in cloud computing. It involves protecting data, applications, and infrastructure from threats.

- **Shared Responsibility Model:** Security is split between the cloud provider (infrastructure security) and the customer (data and application security).
- **Identity and Access Management (IAM):** Controls who can access cloud resources. Uses roles, permissions, and policies.
- **Encryption:** Data should be encrypted both at rest (stored data) and in transit (data being transferred).
- **Firewalls and Security Groups:** Control inbound and outbound traffic to cloud resources.
- **Compliance:** Cloud providers comply with standards like ISO 27001, GDPR, HIPAA to ensure data protection.

8. Scalability and Load Balancing

- **Vertical Scaling (Scale Up):** Increasing the capacity of a single resource (e.g., more CPU or RAM to a server).
- **Horizontal Scaling (Scale Out):** Adding more instances of a resource (e.g., more servers) to handle increased load.
- **Auto-Scaling:** Automatically adjusts the number of running instances based on current demand.
- **Load Balancer:** Distributes incoming network traffic across multiple servers to ensure no single server is overwhelmed. Improves availability and reliability.

9. Cloud Networking

- **Virtual Private Cloud (VPC):** A logically isolated section of the cloud where users can launch resources in a defined virtual network.
- **Subnets:** Subdivisions of a VPC. Public subnets face the internet; private subnets are isolated.
- **DNS (Domain Name System):** Translates domain names to IP addresses. Cloud providers offer managed DNS services.
- **CDN (Content Delivery Network):** A network of servers distributed globally to deliver content to users faster based on their geographic location. Example: AWS CloudFront.

10. Emerging Cloud Concepts

- **Serverless Computing:** Developers write code without managing servers. The cloud provider automatically handles infrastructure. Example: AWS Lambda, Google Cloud Functions.
- **Edge Computing:** Processing data closer to where it is generated (at the edge of the network) rather than in a central cloud. Reduces latency.
- **Cloud-Native Applications:** Applications designed specifically to run in the cloud, using microservices, containers, and CI/CD pipelines.
- **DevOps and CI/CD:** Practices combining development and operations. CI/CD pipelines automate building, testing, and deploying applications in the cloud.

Quick Revision

- **Cloud Computing:** Delivery of computing services over the internet on a pay-as-you-go basis.
- **5 NIST Characteristics:** On-demand self-service, broad network access, resource pooling, rapid elasticity, measured service.
- **Service Models:** IaaS (infrastructure), PaaS (platform), SaaS (software).
- **Deployment Models:** Public, Private, Hybrid, Community, Multi-cloud.
- **Virtualization:** Foundation of cloud; VMs and containers abstract physical hardware.
- **Cloud Storage Types:** Block (databases), Object (files/media), File (shared access).
- **Security:** Shared responsibility model, IAM, encryption at rest and in transit, firewalls.
- **Scalability:** Vertical (scale up), Horizontal (scale out), Auto-scaling, Load balancing.

- **Networking:** VPC, Subnets (public/private), DNS, CDN for faster content delivery.
- **Emerging Concepts:** Serverless (no server management), Edge computing (low latency), Cloud-native, DevOps/CI/CD.